UNIT TEST REPORT

ผลการทดสอบหน่วย (Unit Testing)

ระบบจ่ายยาผู้ป่วยด้วย AR — ทดสอบด้วย Jest (jest-expo preset) + @testing-library/react-native ครอบคลุมตรรกะ AR/QR หลัก, hooks, services, และ utility functions ของแอป

60	60/60	6	97.77%
Test cases ทั้งหมด	ผ่าน (100%)	ไฟล์ทดสอบ	Statement coverage เฉลี่ย

สร้างเมื่อ 14 สิงหาคม 2569 — ผลจริงจากการรัน `npm test -- --verbose --coverage`

การตั้งค่า (Setup)

รายการ	เวอร์ชัน
jest-expo	54.0.17 (Jest 29.7.0)
@testing-library/react-native	14.0.1
babel-preset-expo	54.0.12
Expo SDK ของโปรเจกต์	54

ผลรวม

รายการ	ค่า
Test Suites	6 passed, 6 total
Tests	60 passed, 60 total
เวลารันทั้งหมด	~3 วินาที

รายละเอียด Test Case ทั้งหมด (60 รายการ)

PASS `src/utils/medicationLocatorLogic.test.js` — 16 tests

getBoxRect

- ✓ computes a box from cornerPoints (Android-style result) — 21 ms
- ✓ ignores cornerPoints with fewer than 3 points and falls through — 1 ms
- ✓ falls back to bounds when cornerPoints is absent (iOS-style result) — 2 ms
- ✓ prefers cornerPoints over bounds when both are present — 2 ms
- ✓ falls back to a centered placeholder box when neither geometry is usable — 1 ms
- ✓ falls back when bounds has zero size

matchesSlot

- ✓ returns true when the scanned code equals the patient slot QR
- ✓ returns false for a different code — 1 ms
- ✓ returns false when the patient has no slot QR at all

matchesCassette

- ✓ returns the matching medication for a known cassette QR
- ✓ returns null for an unrecognized QR code — 1 ms
- ✓ returns null when the QR map is empty

isTargetMedication

- ✓ returns true when the matched medication is the current target — 1 ms
- ✓ returns false when the matched medication is a different pending item
- ✓ returns false when there is no current target (e.g. everything already dispensed) — 1 ms
- ✓ returns false when there is no matched medication

PASS `src/utils/qr.test.js` — 6 tests

normalizeQrValue

- ✓ trims surrounding whitespace from a scanned value — 11 ms
- ✓ returns the value unchanged when there is no whitespace to trim — 3 ms
- ✓ returns an empty string for non-string input (e.g. undefined scan result) — 1 ms
- ✓ returns an empty string for null — 1 ms
- ✓ returns an empty string for non-string types like numbers — 1 ms
- ✓ returns an empty string when the value is only whitespace

PASS `src/utils/downloadQrCode.test.js` — 5 tests

downloadQrCode

- ✓ does nothing when the QR ref has not mounted yet — 12 ms

on web

- ✓ builds a temporary download link and clicks it — 5 ms

on native

- ✓ writes the QR image to the cache directory and shares it when sharing is available — 6 ms
- ✓ deletes a pre-existing cached file with the same name before writing the new one — 2 ms
- ✓ does not attempt to share when sharing is unavailable on the device — 12 ms

PASS `src/services/nurseAuthService.test.js` 11 tests

registerNurseAccount

- ✓ rejects an empty full name without calling the API 57 ms
- ✓ rejects an empty username without calling the API 2 ms
- ✓ rejects a password shorter than 6 characters without calling the API 1 ms
- ✓ checks full name and username before password length 2 ms
- ✓ trims whitespace and calls the API with normalized values on success 9 ms
- ✓ propagates the API error (e.g. duplicate username) unchanged 2 ms

loginNurseWithPassword

- ✓ rejects an empty username without calling the API 2 ms
- ✓ rejects an empty password without calling the API 2 ms
- ✓ does not enforce a minimum length on login password (only on registration) 2 ms
- ✓ trims the username and returns a normalized session on success 2 ms
- ✓ propagates the API error (e.g. wrong credentials) unchanged 1 ms

PASS `src/hook/useNurseLogin.test.js` 12 tests

useNurseLogin

- ✓ starts with no session, not loading, and no error 70 ms

handlePasswordLogin

- ✓ sets the session on success 17 ms
- ✓ sets loading true while the request is in flight 10 ms
- ✓ sets the error message and clears loading on failure, without setting a session 7 ms
- ✓ ignores a second call while a login is already in flight (double-submit guard) 8 ms
- ✓ releases the double-submit guard after a failure, allowing a retry 6 ms
- ✓ does not allow a second login attempt after a successful one (lock stays held) 4 ms

handleRegister

- ✓ sets the session on successful registration 4 ms
- ✓ sets the error message on failed registration (e.g. duplicate username) 5 ms
- ✓ is also blocked by the same double-submit guard as login 4 ms

resetLogin

- ✓ clears session, loading, and error back to their initial values 14 ms
- ✓ releases the double-submit guard, allowing login again after a successful session is reset 4 ms

PASS src/hook/useMedicationReminders.test.js 10 tests

useMedicationReminders

- ✓ polls due medications and requests notification permission once on mount (native) 72 ms
- ✓ reflects the fetched list size as dueCount 15 ms
- ✓ surfaces a newly-seen due item as an alert and sends a native notification for it 14 ms
- ✓ does not re-raise an alert for an item already seen in a previous poll 15 ms
- ✓ re-polls fetchDueMedications every 60 seconds 11 ms
- ✓ reschedules daily reminder times every 10 minutes (native only) 12 ms
- ✓ stops polling once unmounted 5 ms
- ✓ skips native notification permission and reminder scheduling on web 4 ms
- ✓ does not crash and leaves state unchanged when the API call fails 7 ms
- ✓ dismissAlert clears the current alert list 11 ms

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	97.77	88.52	93.54	97.56	
hook	96	77.27	88.23	95.52	
useMedicationReminders.js	92.68	72.22	84.61	91.42	25,42-43
useNurseLogin.js	100	100	100	100	
services	100	91.66	100	100	
nurseAuthService.js	100	91.66	100	100	64
utils	100	96.29	100	100	
downloadQrCode.js	100	100	100	100	
medicationLocatorLogic.js	100	94.11	100	100	21
qr.js	100	100	100	100	

Code Coverage

FILE	% STMTS	% BRANCH	% FUNCS	% LINES
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medicationLocatorLogic.js	100	94.11	100	100
qr.js	100	100	100	100

หมายเหตุ: บรรทัดที่ยังไม่ครอบคลุมใน `useMedicationReminders.js` เป็นสาขา Web Notification API (`Notification.permission === 'granted'`) ซึ่งต้องจำลอง browser API เพิ่มเติมนอกขอบเขตที่กำหนดไว้ — ไม่ใช่ตรรกะทางธุรกิจหลักของแอป